

Course Number	0418-455	Course Title	Computer Graphics
Credit Hours	3 (3:1:3)	Course Coordinator	Dr. Mohamed Almulla
Compulsory	No	Prerequisite	0418-301

Textbook

<i>Title:</i>	<i>Computer Graphics – using Open GL</i>
<i>Author:</i>	<i>F. S. Hill, Jr., Stephen M. Kelley</i>
<i>Publisher:</i>	<i>Prentice-Hall, Latest Edition</i>

Major Topics Covered in the Course

	Topics	Hours
1	Introduction: Psychology of human vision; impact of human physiology on graphics hardware and software; graphics systems and models; input /output devices; graphics architectures; camera models.	6
2	Graphical objects and transformations: Primitives, composed objects and data structures associated with them; coordinate systems and frames; geometric objects and transformations; matrix representation of transformations, elements of solid modelling.	10
3	Display algorithms and techniques: Classical computer viewing; positioning of camera; parallel and perspective projections; rendering of scene, Phong's reflection model; polygonal shading, light, color, transparency; clipping of primitives, hidden-surface removal.	11
4	Curves and Surfaces: Parametric polynomial curves, Bezier curves and surfaces, Cubic B-splines, Rendering of curves and surfaces.	8
5	Raster algorithms: Pixel operations, blending colors, logic operations, decomposition of a line, Bresenham's algorithm.	4
6	Exams.	3
	Total.	42

Grading

15-20%	Midterm 1
15-20%	Midterm 2
20-30%	Hmk/ quiz/project
40%	Final

Course Outcomes

Students should be able to:

1. Understand the main areas of computer graphics, graphics software, and graphics hardware.
2. Demonstrate and apply the basic concepts, mathematical models, techniques and algorithms relating to raster graphics.
3. Understand and apply the basic concepts, syntax, and techniques behind the OpenGL graphics library.
4. Demonstrate the basic concepts, mathematical models, techniques and algorithms relating to 2D and 3D modeling and viewing.
5. Demonstrate an understanding of the basic concepts of rendering and visibility of the scene.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1	M					
2	H	M				M
3	L	H				M
4	H	M				M
5	H	M				H